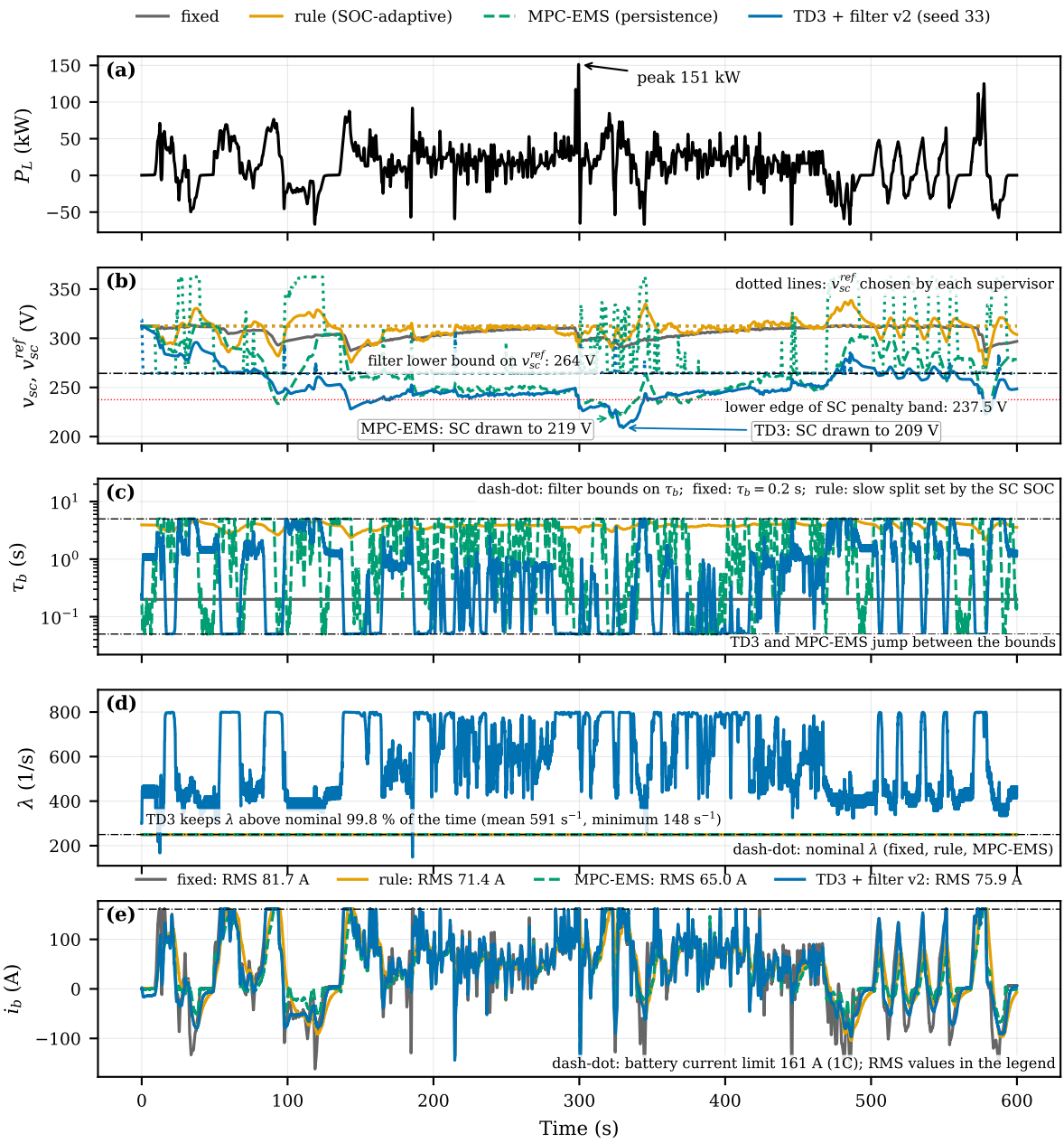

fixed
rule (SOC-adaptive)
MPC-EMS (persistence)
TD3 + filter v2 (seed 33)
(a)
peak 151 kW
P_L (kW)
(b)
dotted lines: v_{sc}^{ref} chosen by each supervisor
filter lower bound on v_{sc}^{ref}: 264 V
lower edge of SC penalty band: 237.5 V
MPC-EMS: SC drawn to 219 V
TD3: SC drawn to 209 V
v_{sc}, v_{sc}^{ref} (V)
(c)
dash-dot: filter bounds on τ_b; fixed: $\tau_b = 0.2$ s; rule: slow split set by the SC SOC
TD3 and MPC-EMS jump between the bounds
τ_b (s)
(d)
TD3 keeps λ above nominal 99.8 % of the time (mean 591 s^{-1}, minimum 148 s^{-1})
dash-dot: nominal λ (fixed, rule, MPC-EMS)
λ (1/s)
fixed: RMS 81.7 A
rule: RMS 71.4 A
MPC-EMS: RMS 65.0 A
TD3 + filter v2: RMS 75.9 A
(e)
dash-dot: battery current limit 161 A (1C); RMS values in the legend
i_b (A)
Time (s)